

HACKATHON PROJECT

AttendX

Autonomous Attendance & Proxy Audit Engine

An engine that checks students in by face, catches proxy marks, parses messy leave notes, and fires real warning letters — while a professor keeps the final call, inside Notion.



Facial verification • LLM leave parsing • Notion API • Twilio/Meta API • PDF generation

THE PROBLEM

Attendance day is a 4-hour weekly slog



Manual marking

Roll calls, spreadsheets, and re-typing into registers eat hours every single week.



Undetected proxy marks

Friends mark "present" for absent students — no way to catch it at scale.



Messy leave notes

Excuses arrive as scattered WhatsApp texts and emails, impossible to standardize.



Manual warning letters

Drafting and sending defaulter notices to parents is slow, inconsistent, and often skipped.

Result: proxy attendance goes unnoticed and defaulters slip through the cracks.

THE SOLUTION

One engine. Notion as the operations center.

AttendX automates check-in → AI verification → human review → real notice dispatch, end to end — for any college department.



AI does the verification

Face matching scores every kiosk check-in; an LLM classifies messy leave notes into structured fields.



Human stays in control

Proxy flags and defaulter actions wait in Notion until a professor approves them.



Real-world action

Approved defaulters get an actual WhatsApp/email notice with a generated PDF letter — not a simulated one.



Everything is logged

Every check-in, leave submission, and defaulter job writes a timestamped row to the Run Log.

HOW IT WORKS

The pipeline, end to end



Branching logic: a clean match (>85%) auto-marks "Present." Mismatches and defaulters route straight to the Review & Action Desk for a human decision.

Three databases, one control workspace



Student Master & Register

- Roll No, Name, Parent Phone
- Total Classes · Attended
- Attendance % (formula)
- Profile Photo — gallery view



Review & Action Desk

- Filtered to Status = Needs Review
- Proxy Flags · Leave Requests
- Defaulter Actions
- Human-in-the-loop hub



System Run Log

- Read-only, written via API
- Timestamp · Trigger Event
- Payload Summary · Status
- Execution Time (ms)

Even offline, this workspace reads like a real operations dashboard — not a raw data dump.

Where AI actually earns its place

Two real AI calls in the pipeline — both need judgment a fixed rule can't provide.



1. Facial Verification

Live kiosk snapshot → confidence score

- Compares the live kiosk snapshot against the baseline roster photo
- Assigns a similarity confidence score to every check-in
- High confidence (>85%) marks attendance immediately
- Low confidence routes straight to the Proxy Flag Desk for human review



2. Messy Leave Parser

Unstructured excuse → structured record

- Reads messy leave notes and emails ("bike broke down near highway...")
- Extracts category, duration, and a short reason summary
- Assesses urgency automatically
- Attaches the structured data directly to the student's daily register entry

A professor always makes the final call



Proxy Flags

Baseline photo vs. kiosk snapshot side-by-side, with the AI confidence score. Professor clicks Confirm Present or Mark Absent & Penalty.



Messy Leave Requests

AI-structured fields — Category, Duration, Reason Summary. Professor reviews and clicks Approve Leave.



Defaulter Actions

Students under 75% attendance. A single checkbox: Dispatch Parent Warning Notice.



Nothing is auto-sent. Every flagged case sits in Notion until a professor makes the call.

Beyond Notion: real notices, not simulations



WhatsApp / SMS Alerting

- Real-time automated alerts to parents on unexcused absence
- Fires the moment a professor approves a defaulter notice
- Sent via Twilio / Meta WhatsApp Business API
- No manual dialing, typing, or forwarding by a human



Official Notice PDF Generation

- Backend compiles attendance data into a formatted PDF
- Includes college letterhead, calculated shortage, and regulation citations
- Emailed directly to the department head
- A real signed document, not a placeholder message

Tech stack



Backend Engine

Runs face matching, leave classification, and attendance calculations



Facial Recognition Model

Matches the live kiosk snapshot to the baseline roster photo, returns a confidence score



LLM Leave Classifier

Extracts structured fields from messy leave notes and emails



Notion API

3 databases — Student Register, Review Desk, Run Log — the control workspace



Twilio / Meta WhatsApp API

Dispatches real-time parent alerts



PDF Generation Engine

Compiles formatted warning letters with letterhead & regulation citations

Why this wins the track



Engine-first

Delete the backend repo, and live scanning, AI verification, notifications, and audit logging all stop working.



Notion as an operations center

Even with the server offline, a department head can open Notion and see an organized, human-readable dashboard.



Zero busywork for humans

Normal operations are 100% autonomous — a professor only touches Notion when an anomaly or high-stakes decision occurs.

Live demo flow



1. Show the Notion workspace

Student Register + Run Log, live and human-readable.



3. Trigger a mismatched check-in

Low confidence — routes straight to the Proxy Flag Desk.



5. Approve a defaulter action

Real WhatsApp alert + PDF letter fires to parent and HOD.



2. Trigger a clean check-in

High-confidence match — student auto-marked Present instantly.



4. Review in Notion

Professor confirms/flags the proxy case and approves a leave note.



6. Show the new Run Log row

Real timestamp — "turn off the backend, Notion still stands alone."



Thank You

AttendX — Autonomous Attendance & Proxy Audit Engine

Questions? Let's see it live.